



Draft genome sequence of a multidrug-resistant novel candidate *Pseudomonas* sp. NCCP-436^T isolated from faeces of a bovine host in Pakistan

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ABSTRACT

Objectives: Here we describe the first draft genome analysis of a CRISPR-carrying, multidrug-resistant, candidate novel *Pseudomonas* sp. NCCP-436^T isolated from faeces of a neonatal diarrhoeic calf.

Methods: The genome of strain NCCP-436^T was sequenced using an Illumina NovaSeq PE150 platform and analysed using various bioinformatic tools. The virulence factors and resistome were identified using PATRIC and CARD servers, while CGView Server was used to construct a circular genome map. Antimicrobial susceptibility was determined by the disk diffusion technique.

Results: The draft genome of strain NCCP-436^T contains 43 contigs with a total genome size of 3,683,517 bp (61.4% GC content). There are 3,452 predicted genes, including 60 tRNAs, 7 rRNAs and 12 sRNAs. CRISPR analysis revealed two CRISPR arrays with lengths of 1103 bp and 867 bp. Strain NCCP-436^T was highly resistant to fluoroquinolone, β -lactam, cephalosporin, aminoglycoside, penicillin, rifamycin, macrolide, glycopeptide, trimethoprim/sulfonamide and tetracycline antibiotic classes. Additionally, 22 antibiotic resistance genes, 313 virulence genes and 253 pathogen–host interactor genes were predicted. Comparison of the average nucleotide identity and digital DNA–DNA hybridisation values with the closely-related strain *Pseudomonas khazarica* (TBZ2) was found to be 82.08% and 34.90%, respectively, illustrating strain NCCP-436^T as a potentially new species of *Pseudomonas*.

Conclusion: Substantial number of antibiotic resistance and virulence genes and homology with human pathogens were predicted, exposing the pathogenic and zoonotic potential of strain NCCP-436^T to public health. These findings may be used to better understand the genomic epidemiological features and drug resistance mechanisms of pathogenic *Pseudomonas* spp. in Pakistan.

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Pakistan has a diverse genetic reserve in farm animals, well adapted to local environmental conditions. Pathogenic bacteria excreted in animal faeces have been found in nature and demonstrated high resistance to a large number of available antibiotics [1]. As many animals and their products are traded around the

world, antimicrobial resistance in one country is an issue for all countries [2]. In this context, the main aim of the present work was to assess the prevalence of bacterial pathogens and the antibiotic resistance potential in the faecal ecosystem of food animals in Pakistan. This study reports the first draft genome sequence analysis of a CRISPR-carrying, multidrug-resistant (MDR), novel candidate *Pseudomonas* sp. NCCP-436^T recovered from the faecal matter

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of a host animal and thus serves as a foundation for future epidemiological and genetic studies of the infectious pathogens.

Members of the genus *Pseudomonas*, one of the most complex Gram-negative bacterial genus with more than 396 species, are able to grow in multiple environmental niches [3] and display tremendous variation in genome size [4,5], ranging from 3.02 Mb to 7.38 Mb in *Pseudomonas caeni* and *Pseudomonas saponiphila*, respectively. Many human and agricultural diseases associated with opportunistic infections have been attributed to *Pseudomonas* species [2]. Although indiscriminate antibiotic use has contributed to the emergence of MDR *Pseudomonas* spp. worldwide [6], there is a scarcity of whole-genome sequencing (WGS) data from animal hosts. As a result, draft genome findings of *Pseudomonas* strain NCCP-436^T, isolated from a faecal sample of a neonatal calf in 2018 from Pakistan and showing resistance to various antibiotic classes, are presented here for the first time to better understand the pathogen's evolution and epidemiology in order to execute accelerated infection control programmes.

Tryptic soy agar and nutrient agar media were used to isolate strain NCCP-436^T at a temperature of 30°C. Strain NCCP-436^T was purified by repeated streaking and was identified by 16S rRNA gene sequencing subjected to BLAST search using the Ez-Taxon server (<http://eztaxon-e.ezbiocloud.net>). The disk diffusion technique was performed to assess antimicrobial susceptibility according to Clinical and Laboratory Standards Institute (CLSI) guidelines [7]. Strain NCCP-436^T was subjected to WGS using an Illumina NovaSeq PE150 platform (Illumina Inc.) at Novogene Co. (Beijing, China) and was analysed using various bioinformatics tools. The sequence reads of each data set were filtered and high-quality paired-end reads were de novo assembled using SPAdes v.3.13 [8] (<http://cab.spbu.ru/software/spades/>), while completeness and contamination of the genome were calculated using CheckM [9]. Gene annotation was predicted using online databases including Kyoto Encyclopedia of Genes and Genomes (KEGG) (<https://www.genome.jp/kegg>), Clusters of Orthologous Groups (COG) (<https://www.ncbi.nlm.nih.gov/research/cog-project>), Gene Ontology (GO) (<http://geneontology.org>), Rapid Annotation using Subsystems Technology (RAST) (<http://rast.nmpdr.org/rast.cgi>) and Carbohydrate-Active enZymes (CAZy) (<http://www.cazy.org>). The TMHMM Program (<http://www.cbs.dtu.dk/services/TMHMM>) was used to investigate signal peptides, transmembrane helices and secretory protein-encoding genes. In particular, the Pathosystems Resource Integration Center (PATRIC) (<https://www.patricbrc.org>) and Comprehensive Antibiotic Resistance Database (CARD) (<https://card.mcmaster.ca>) databases were searched for identification of the resistome and virulence factors in the genome of strain NCCP-436^T. CRISPRfinder (<https://crisprcas.i2bc.paris-saclay.fr>) and PHASTER databases were used to predict clustered regularly interspaced short palindromic repeats (CRISPR) and prophage regions. The digital DNA–DNA hybridisation (digi-DDH) and average nucleotide identity (ANI) values for *Pseudomonas* sp. NCCP-436^T compared with the genomes of closely-related validly named type strains were determined using the Genome-to-Genome Distance Calculation [10] (GGDC 2.0) (<http://ggdc.dsmz.de>) and the orthoANI calculator on EzTaxon-e server [11] (<http://eztaxon-e.ezbiocloud.net>). Finally, a circular map of the genome of strain NCCP-436^T was constructed (Fig. 1) using CGView [12] (<http://cgview.ca>).

The draft genome of *Pseudomonas* strain NCCP-436^T contains 43 contigs with a total genome size of 3,683,517 bp, 240 × coverage and a GC content of 61.4%. There are 3,452 predicted genes, including 60 tRNAs, 7 rRNAs (5 copies of 5S and single copies each of 16S and 23S) and 12 sRNA as well as 3 genomic islands. CRISPR analyses revealed two CRISPR arrays with lengths of 1103 bp and 867 bp on the positive strand of scaffold 1, while functional annotation demonstrated the absence of prophage-related genes.

Repetitive DNA analysis confirmed the existence of 57 tandemly repetitive DNA sequences, while minisatellite DNA and microsatellite DNA were located in numbers of 41 and 5, respectively. A total of 72 genes encoding carbohydrate-active enzymes were identified, while determination of secondary metabolites revealed 65 secondary metabolites genes distributed on two gene clusters, i.e. non-ribosomal peptide synthetase (NRPS) cluster and β -lactone cluster. Antibiotic susceptibility testing revealed that strain NCCP-436^T was highly resistant to 15 drugs belonging to the fluoroquinolones (ciprofloxacin, bacitracin and moxifloxacin), β -lactams (amoxicillin/clavulanic acid), third-generation cephalosporins (cef-tazidime, cefoperazone and cefotaxime), aminoglycosides (gentamicin and amikacin), penicillins (ampicillin), rifamycins (rifampicin), macrolides (erythromycin), glycopeptides (vancomycin), trimethoprim/sulfonamide (sulfamethoxazole/trimethoprim) and tetracyclines (tetracycline) while being sensitive to carbapenems (imipenem). Minimum inhibitory concentrations (MICs) were determined to ciprofloxacin (30 mg/L), sulfamethoxazole (10 mg/L), amoxicillin (20 mg/L), ampicillin (10 mg/L), tetracycline (3 mg/L), levofloxacin (1 mg/L) and ofloxacin (0 mg/L) antibiotics (Sigma-Aldrich, Germany) [13]. As evident from the phenotype, 22 antibiotic resistance genes and 54 antibiotic resistance gene clusters and efflux pump complexes were predicted that confer resistance to multiple antibiotic classes via molecular bypass (Supplementary Table S2). Additionally, 313 virulence genes and 253 pathogen–host interactor genes were also found in the genome of *Pseudomonas* sp. NCCP-436^T as illustrated in Fig. 1. The most common genes were identified as resistance–nodulation–cell division (RND)-type multidrug efflux pump genes for resistance to chloramphenicol and fluoroquinolones (i.e. *mexE*). Two other multidrug resistance efflux pump genes (*mexA* and *mexV*) were associated with aminoglycoside, tigecycline, fluoroquinolone and β -lactam resistance. Further detailed information on all of the antibiotic resistance genes in relation to their respective drug classes are listed in Supplementary Table S2. Harboring the substantial predicted ARGs and homology to human pathogens, strain *Pseudomonas* sp. NCCP-436^T may pose a potential public health risk, thus in One-Health perspective, a public health action plan is urgently needed to combat the antimicrobial resistance phenomenon in faecal ecology of neonatal diarrhoeic animals in Pakistan.

To establish the novel candidate status of strain NCCP-436^T, a phylogenomic tree was constructed on the basis of whole-genome sequences of closely-related validly named type strains of the genus *Pseudomonas* (Supplementary Fig. S1). Similarly, a phylogenetic analysis was also performed on the basis of 16S rRNA gene sequences and a tree was constructed using three algorithms, namely neighbour-joining, maximum likelihood and maximum parsimony (Supplementary Fig. S2). These phylogenetic trees depicted the monoclade position of isolate NCCP-436^T, suggesting it as a new member of the genus *Pseudomonas*. Moreover, genome-assisted comparison of ANI and digi-DDH values with the closely-related strains were found to be <95–96% and <70%, respectively (Supplementary Table S1), which are below the threshold value for species delineation of isolate NCCP-436^T [14,15]. These findings show that strain NCCP-436^T could potentially be a new species of the *Pseudomonas* genus from Pakistan, although further polyphasic taxonomic characterisation is needed.

GenBank accession numbers

The DDBJ/EMBL/GenBank accession number for the 16S rRNA gene sequence of *Pseudomonas* NCCP-436^T is **LC488921**. The BioProject, BioSample and whole-genome sequences of strain NCCP-436^T have been deposited in DDBJ/EMBL/GenBank under the accession numbers **PRJDB11788**, **SAMD00328564** and **BPLZ01000001–BPLZ01000043**.

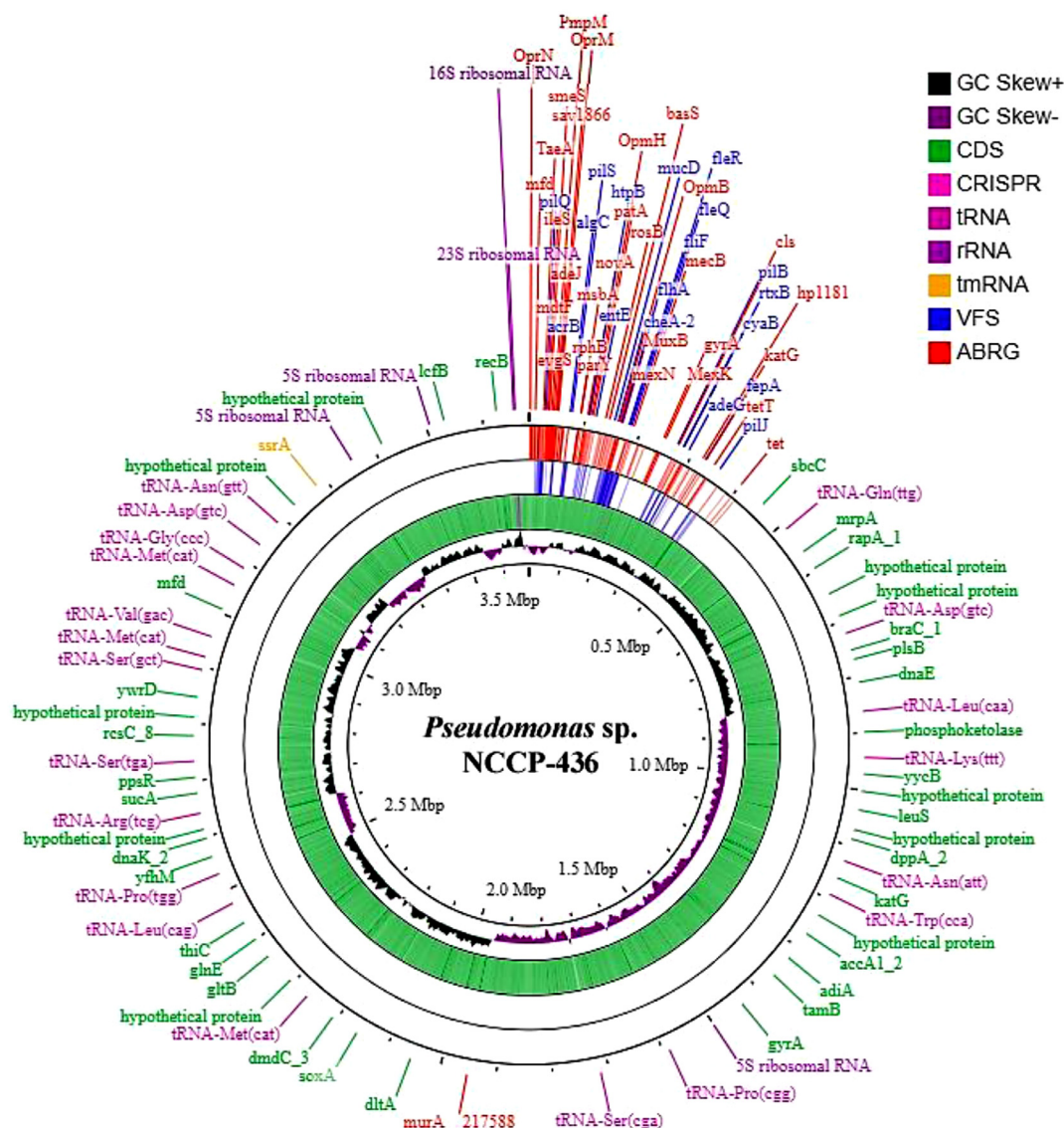


Fig. 1. Graphical circular genome map of *Pseudomonas* sp. NCCP-436T constructed using CGView [12] (<http://cgview.ca>). From outside to inside, rings 1 and 2 are annotated antibiotic resistance genes (ABRG) and virulence factor genes (VFS), respectively. Ring 3 represents coding and non-coding sequences (including tRNA, rRNA and sRNA). The GC skew pattern is represented by the inner most ring, with purple indicating negative values and black indicating positive values.

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Competing interests

None declared.

Ethical approval

Not required.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2021.08.011](https://doi.org/10.1016/j.jgar.2021.08.011).

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